📦 Unit 9: Volume Practice — ANSWER KEY

Cylinders · Cones · Spheres · Composite Solids · NC.8.G.9

Name: _____ Date: _____ Period: _____

Cylinder: $V = \pi r^2 h$ **Cone: $V = \frac{1}{3}\pi r^2 h$** **Sphere: $V = \frac{4}{3}\pi r^3$**

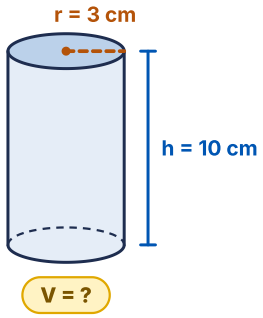
π ≈ 3.14 · Always use cubic units · Remember: radius = diameter ÷ 2

💡 **Tip:** A cone is exactly ⅓ of a cylinder with the same dimensions. When you see "diameter," divide by 2 to get radius first!

🥫 Cylinder Problems

Problem V1

A cylinder has radius 3 cm and height 10 cm. What is the volume? (Use π ≈ 3.14)

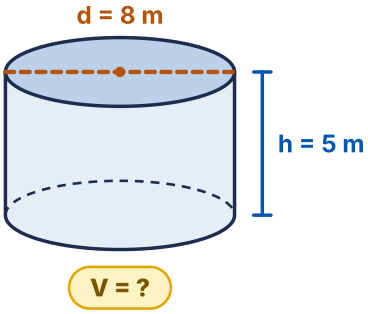


✓ **282.6 cm³**
○ 94.2 cm³
○ 188.4 cm³
○ 1130.4 cm³

Solution: $V = \pi(3)^2(10) = \pi(9)(10) = 90\pi \approx 282.6$ cm³

Problem V4

A cylinder has diameter 8 m and height 5 m. What is the volume?

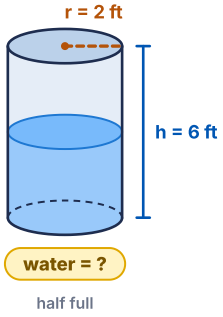


✓ **80π m³**
○ 160π m³
○ 320π m³
○ 40π m³

Solution: Diameter = 8, so r = 4. $V = \pi(4)^2(5) = \pi(16)(5) = 80\pi \approx 251.2$ m³

Problem V10

A cylindrical water tank (r = 2 ft, h = 6 ft) is half full. How much water is inside?



✓ **12π ft³**
○ 24π ft³
○ 6π ft³
○ 8π ft³

Solution: Full: $V = \pi(2)^2(6) = 24\pi$. Half full: $24\pi \div 2 = 12\pi$ ft³

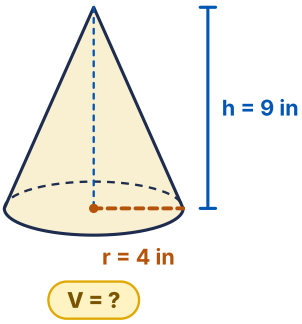
🍦 Cone Problems

Problem V2

A cone has radius 4 in and height 9 in. What is the volume?

- ○ 64π in³
- ✔ **48π in³**
- ○ 452.16 in³
- ○ 36π in³

Solution: V = ⅓π(4)²(9) = ⅓π(16)(9) = ⅓(144π) = 48π ≈ 150.72 in³

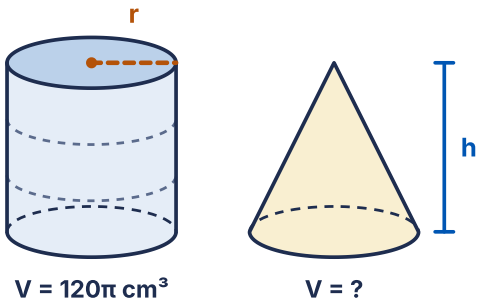


Problem V5

A cone and cylinder have the same radius and height. The cylinder's volume is 120π cm³. What is the cone's volume?

- ✔ **40π cm³**
- ○ 60π cm³
- ○ 360π cm³
- ○ 80π cm³

Solution: A cone is ⅓ of a cylinder! 120π ÷ 3 = 40π cm³

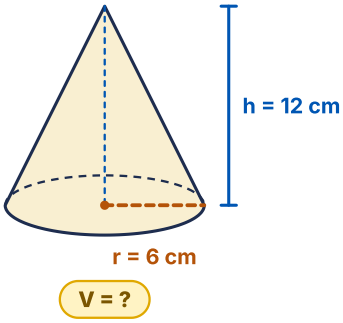


Problem V7

A cone has radius 6 cm and height 12 cm. What is the volume? (Use π ≈ 3.14)

- ○ 432π cm³
- ✔ **452.16 cm³**
- ○ 1356.48 cm³
- ○ 226.08 cm³

Solution: V = ⅓π(6)²(12) = ⅓π(36)(12) = ⅓(432π) = 144π ≈ 452.16 cm³



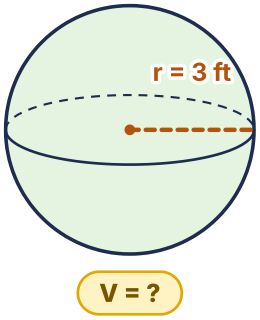
🏀 Sphere Problems

Problem V3

A sphere has radius 3 ft. What is the volume?

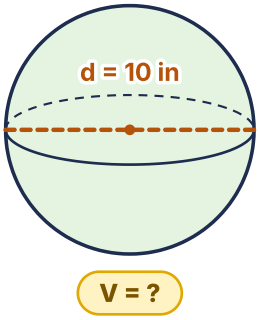
- ✔ **36π ft³**
- ○ 27π ft³
- ○ 84.78 ft³
- ○ 108π ft³

Solution: V = ⁴⁄₃π(3)³ = ⁴⁄₃π(27) = 36π ≈ 113.0 ft³



Problem V6

A sphere has diameter 10 in. What is the volume?



✓ **500π/3 in³**

○ 4000π/3 in³

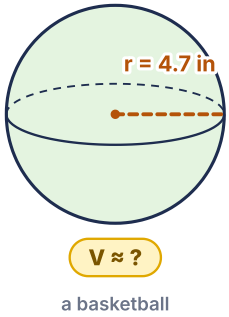
○ 250π in³

○ 125π in³

Solution: Diameter = 10, r = 5. V = ⁴⁄₃π(5)³ = ⁴⁄₃π(125) = 500π/3 ≈ 523.3 in³

Problem V8

A basketball has radius 4.7 in. Approximately what is its volume? (Use π ≈ 3.14)



a basketball

✓ **434.7 in³**

○ 138.2 in³

○ 275.5 in³

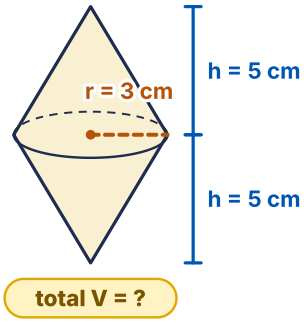
○ 207.1 in³

Solution: V = ⁴⁄₃π(4.7)³ = ⁴⁄₃(3.14)(103.823) ≈ 434.7 in³

📦 Mixed & Composite

Problem V9

Two identical cones are placed base-to-base. Each cone has r = 3 cm and h = 5 cm. What is the total volume?



✓ **30π cm³**

○ 60π cm³

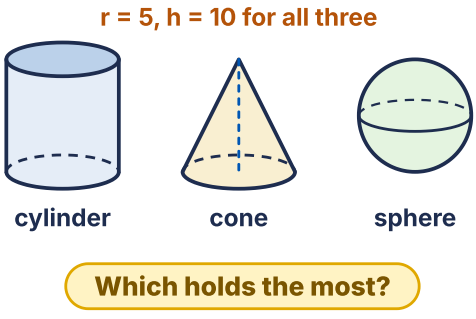
○ 15π cm³

○ 90π cm³

Solution: One cone: V = ⅓π(3)²(5) = 15π. Two cones: 2(15π) = 30π cm³

Problem V11

Which shape has the greatest volume? r = 5, h = 10 for all.



✓ **Cylinder**

○ Cone

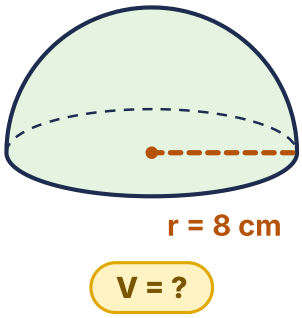
○ Sphere

○ They're all equal

Solution: Cylinder: π(25)(10) = 250π. Cone: ⅓(250π) ≈ 83.3π. Sphere: ⁴⁄₃π(125) ≈ 166.7π. Cylinder wins!

Problem V12

A hemisphere (half sphere) has radius 8 cm. What is its volume?



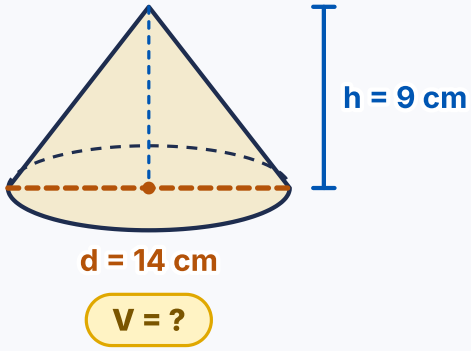
- ✔ **1024π/3 cm³**
- ○ 2048π/3 cm³
- ○ 256π cm³
- ○ 512π cm³

Solution: Full sphere: $\frac{4}{3}\pi(8)^3 = 2048\pi/3$. Hemisphere = half: $2048\pi/6 = 1024\pi/3$ cm³

📝 Worked Examples

Study these step-by-step solutions before attempting the problems above.

Volume of a cone: diameter = 14 cm, height = 9 cm



diameter 14 → radius 7

Step 1: Identify r — diameter = 14, so r = 7

$r = 7,\ h = 9$

Step 2: Write the formula

$V = \frac{1}{3}\pi r^2 h$

Step 3: Substitute

$V = \frac{1}{3}\pi(7^2)(9) = \frac{1}{3}\pi(49)(9)$

Step 4: Multiply inside

$V = \frac{1}{3}(441\pi)$

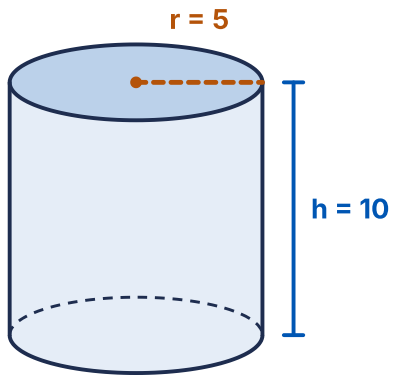
Step 5: Divide by 3

$\mathbf{V = 147\pi \approx 461.6\ cm^3}$ ✓

🔍 Common Mistakes — Can You Spot the Error?

Each problem has a mistake. Identify what went wrong.

Cylinder: r = 5, h = 10

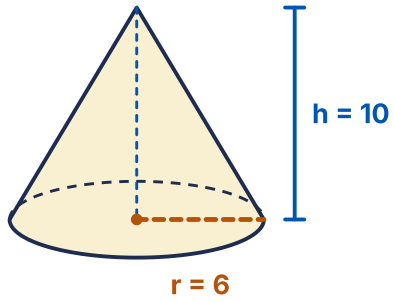


✗ **Student:** V = π(5)(10) = 50π

✓ **Fix:** Forgot to SQUARE the radius! V = π(5²)(10) = π(25)(10) = 250π

💡 *The formula is πr²h — that ² is critical!*

Cone: r = 6, h = 10

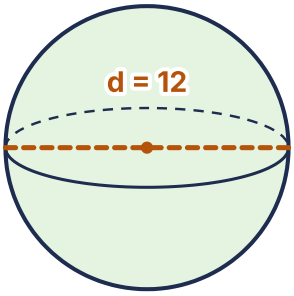


✗ **Student:** V = π(6²)(10) = 360π

✓ **Fix:** That's the CYLINDER formula! Cones need ⅓. V = ⅓π(36)(10) = 120π

💡 *A cone is always ⅓ of the matching cylinder.*

Sphere: diameter = 12



the 12 goes all the way across

✗ **Student:** $V = \frac{4}{3}\pi(12)^3 = \frac{4}{3}\pi(1728) = 2304\pi$

✔ **Fix:** Used DIAMETER instead of radius! r = 6, not 12. $V = \frac{4}{3}\pi(6)^3 = \frac{4}{3}\pi(216) = 288\pi$

💡 *Always check: did the problem give diameter or radius? Diameter ÷ 2 = radius.*